

Dexter Fernandes

Bristol • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns •
<https://github.com/Dexter-Fernandes>

Machine Vision Engineer

AI Software Engineer with 3+ years delivering real-time, safety-critical perception systems on edge hardware. Deep expertise in deploying vision models using PyTorch, C++17, and ROS2, with hands-on FP16 and INT8 quantisation on Intel VPUs via OpenVINO under strict latency, power and reliability constraints. Strong background translating research into deterministic, production-ready deployments through CI, profiling, and on-site validation. Experienced collaborating across AI, robotics, and hardware teams to ship resilient edge AI systems operating 24/7 in harsh, unstructured environments.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed and deployed production computer vision applications for real-time video analytics on edge devices delivering deterministic 24/7 operation in safety-critical oil-rig environments.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.
- Designed and validated deterministic system behaviour through on-site field testing, performing root-cause analysis of model failures and reducing crash frequency by 67% in live video pipelines.
- Refactored legacy monolith into Kafka-driven, Dockerised microservices, improving fault isolation and increasing MTBF by 35% across deployments.
- Owned end-to-end data pipelines, including automated data collection, augmentation, and labelling, increasing dataset diversity by 40% and reducing manual labelling effort by 60%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Validated deterministic model performance under extreme field conditions at a British Petroleum site, ensuring reliability across 24/7 operations.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led technical delivery of real-time computer vision applications for object detection, pose estimation, and tracking, achieving sub-50 ms inference latency and reducing false positives by 33% on embedded hardware.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Built automated labelling, dataset versioning, and performance-tracking pipelines, repurposing detection models for annotation and cutting manual labelling effort by 80%, accelerating iteration cycles.
- Collaborated with hardware, software, and QA teams to integrate AI modules into production systems, mentoring engineers on optimisation and deployment practices to improve delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes and interfaces integrating perception outputs into a broader robotic system, managing sensor topics, tf2 frames, and message synchronisation for downstream localisation and navigation.
- Integrated perception pipelines into ROS-based robotic workflows, validating end-to-end data flow from sensors to SLAM and navigation components in Gazebo prior to deployment.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 3D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full application using Docker to ensure reproducibility, version control, and consistent deployment across development environments.

SKILLS

- **Perception & SLAM:** 3D Feature Extraction, Convolutional Neural Networks, Data Augmentation, Eigen, LiDAR SLAM, NumPy, OpenCV, Point Cloud Library, ROS2 image_transport, SAM, Scikit-Image, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, Docker, DVC (Data Version Control), Git, Intel VPUs, Linux, Microsoft Azure, MLFlow
- **Programming & Development:** Bash, C++17, CMake, colcon, Gazebo (Harmonic), Nav2, Pandas, PostgreSQL, Python, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)
- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT